

## **CO5-AT2-Q1:**

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**Course Name: DAA for Complexity Analysis**

### **Approximation Bin Packing Problem**

### **Problem Understanding**

Bin Packing problem focuses on placing items into minimum number of bins where each bin has limited capacity. The objective is to reduce unused space and optimize storage.

First-Fit Decreasing is an approximation algorithm that sorts items in decreasing order and places each item into the first suitable bin.

### **Algorithm Used**

First-Fit Decreasing (Greedy Approximation)

### **Constraints**

- Bin capacity should not be exceeded.
- Every item must be assigned to a bin.
- Number of bins should be minimized.

### **Solution Approach**

Steps:

Sort all items in decreasing order.

Create empty bins.

Take each item one by one.

Check existing bins.

Place item in first bin where it fits.

If no bin is available, create a new bin.

### **Correctness**

The algorithm places every item into a valid bin without exceeding capacity. Sorting improves the probability of efficient packing.

## Complexity Analysis

Sorting:

$O(n \log n)$

Placement:

$O(n^2)$

Overall:

$O(n^2)$

## Conclusion

First-Fit Decreasing provides an efficient approximation solution for Bin Packing. It is useful in storage allocation, scheduling, and resource management problems.